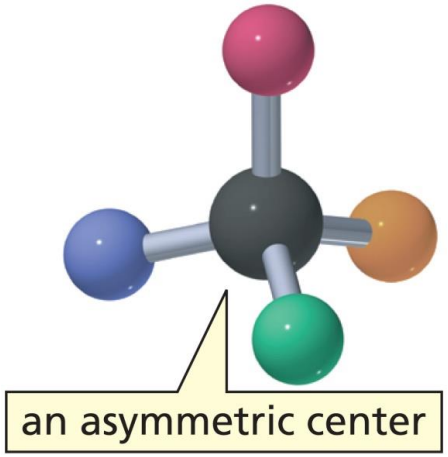


OPTICAL ISOMERISM

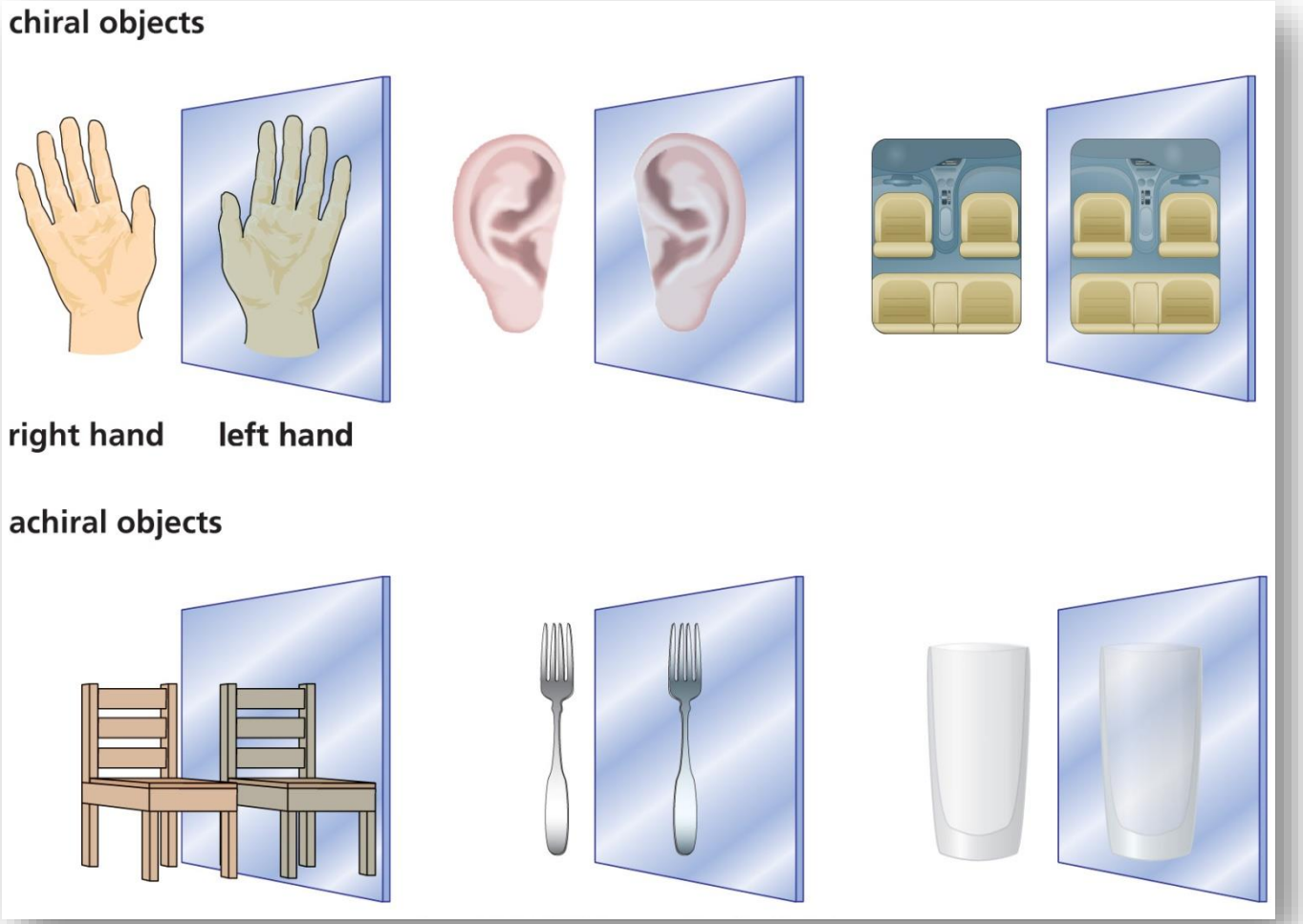
Optical Isomers – with mirror images

- are a type of stereoisomer that have the same molecular formula and same connectivity but differ in the three-dimensional arrangement of atoms such that they are non-superimposable mirror images of each other.
    - One isomer rotates light clockwise (**dextrorotatory**, +)
    - The other rotates light counter-clockwise (**levorotatory**, −).
    - Optical isomers usually arise from the presence of a **chiral center**— a carbon atom bonded to four different groups.



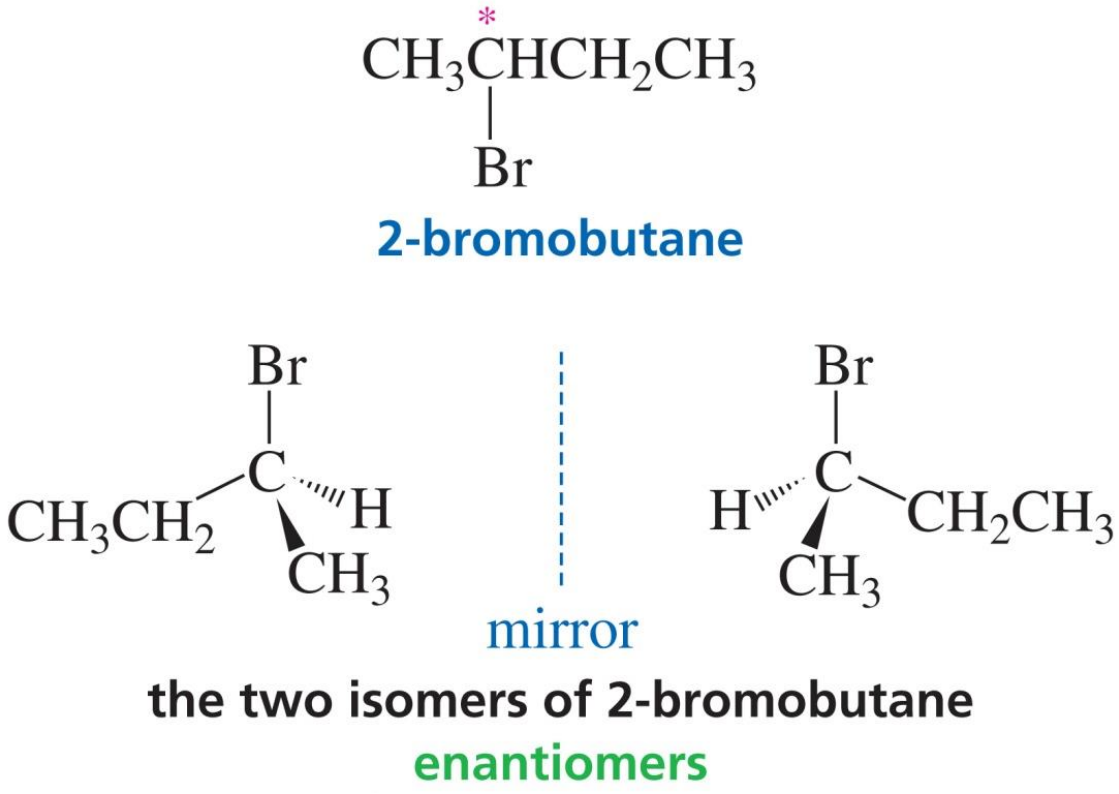
Chirality

- **Chiral objects** are objects with left-handed and right-handed forms
- **Achiral objects** - objects that have superimposable mirror images
- **Nonsuperimposable mirror images** - a mirror image that is not the same as the image itself - chiral objects have nonsuperimposable mirror images

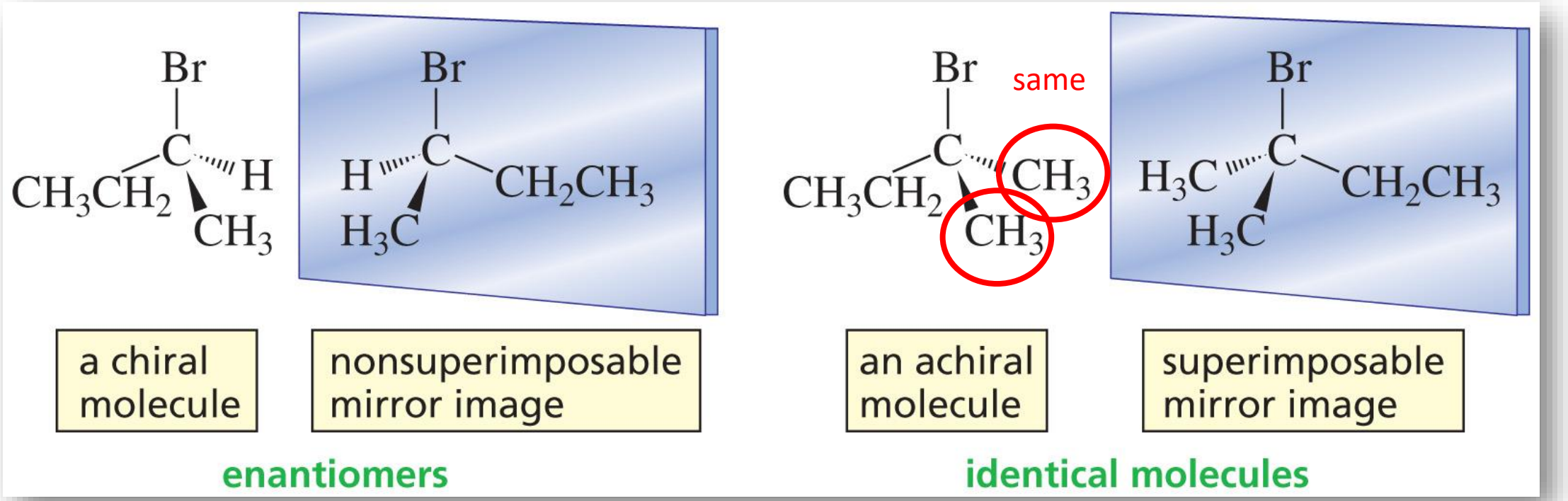


Chiral Molecules

- Molecules with **one** chiral center would have **2** enantiomers
- Chiral molecules would have 2^n enantiomers (where n is the number of chiral centers)



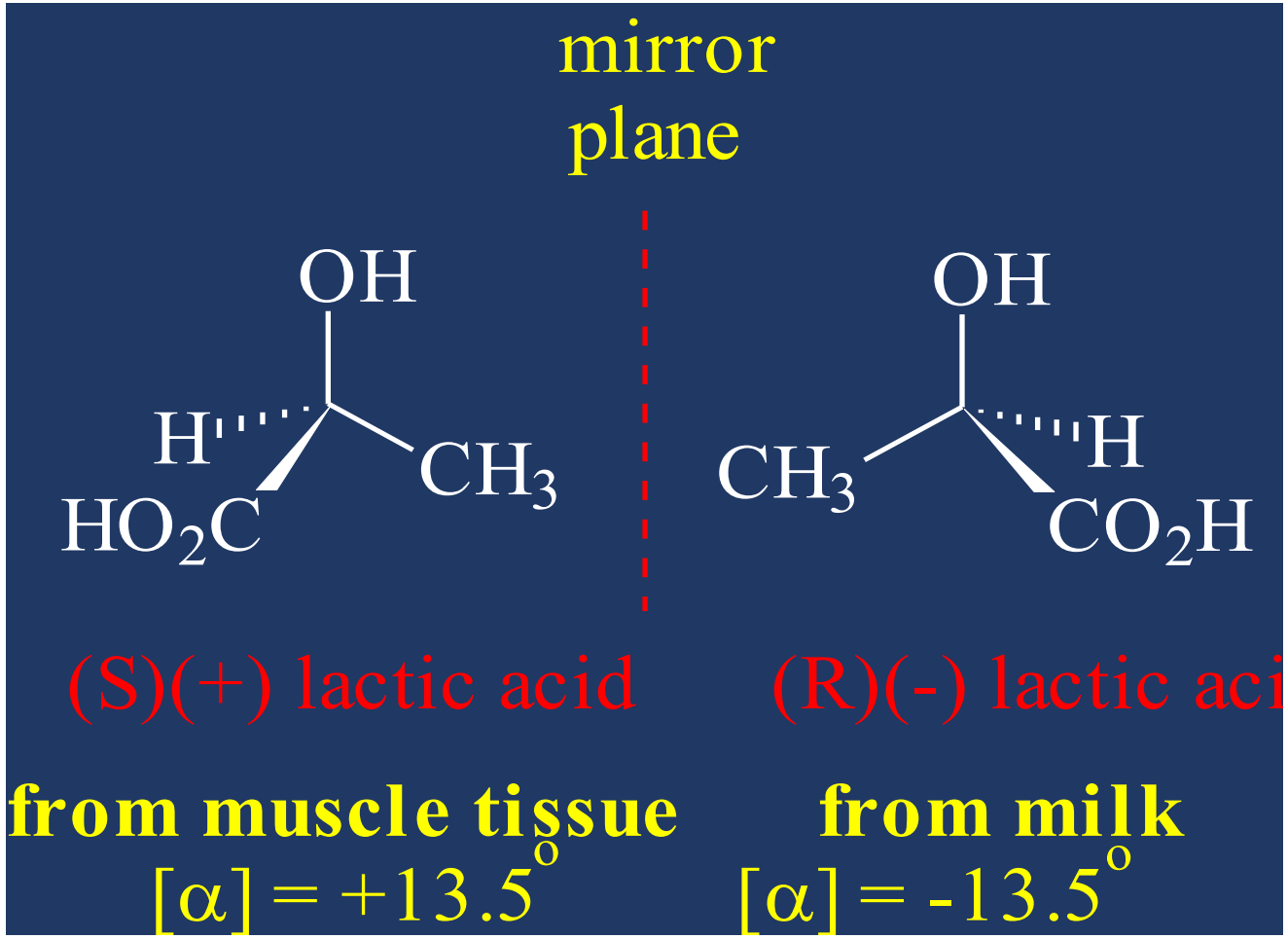
Chiral vs Achiral Molecules



Stereoisomers

- **Stereoisomers** are either enantiomers or diastereomers.
  - **Enantiomers**: are a pair of optical isomers that are non-superimposable mirror images of each other. They have:
    - The same molecular formula
    - The same connectivity
    - The same physical properties (melting point, boiling point, density)
    - Difference: They rotate plane-polarized light in opposite directions (one dextro, other levo)
  - **Diastereomers:** are chiral molecules but are not mirror images of each other and are not superimposable.
    - Unlike enantiomers, diastereomers have noticeably different physical and chemical properties, such as boiling point, solubility, and reactivity

Enantiomers - non-superimposible mirror images



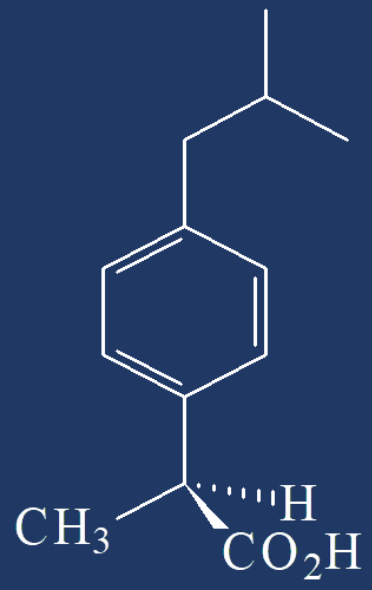
EXAMPLES

Pseudophed

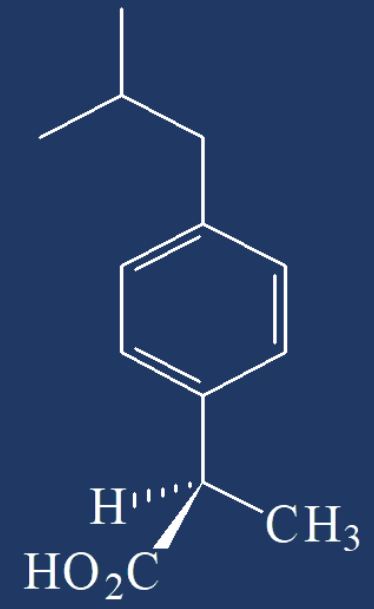


"desoxyephedrine"
(methamphetamine

l-(R) - Vicks
d-(S)- "Meth"



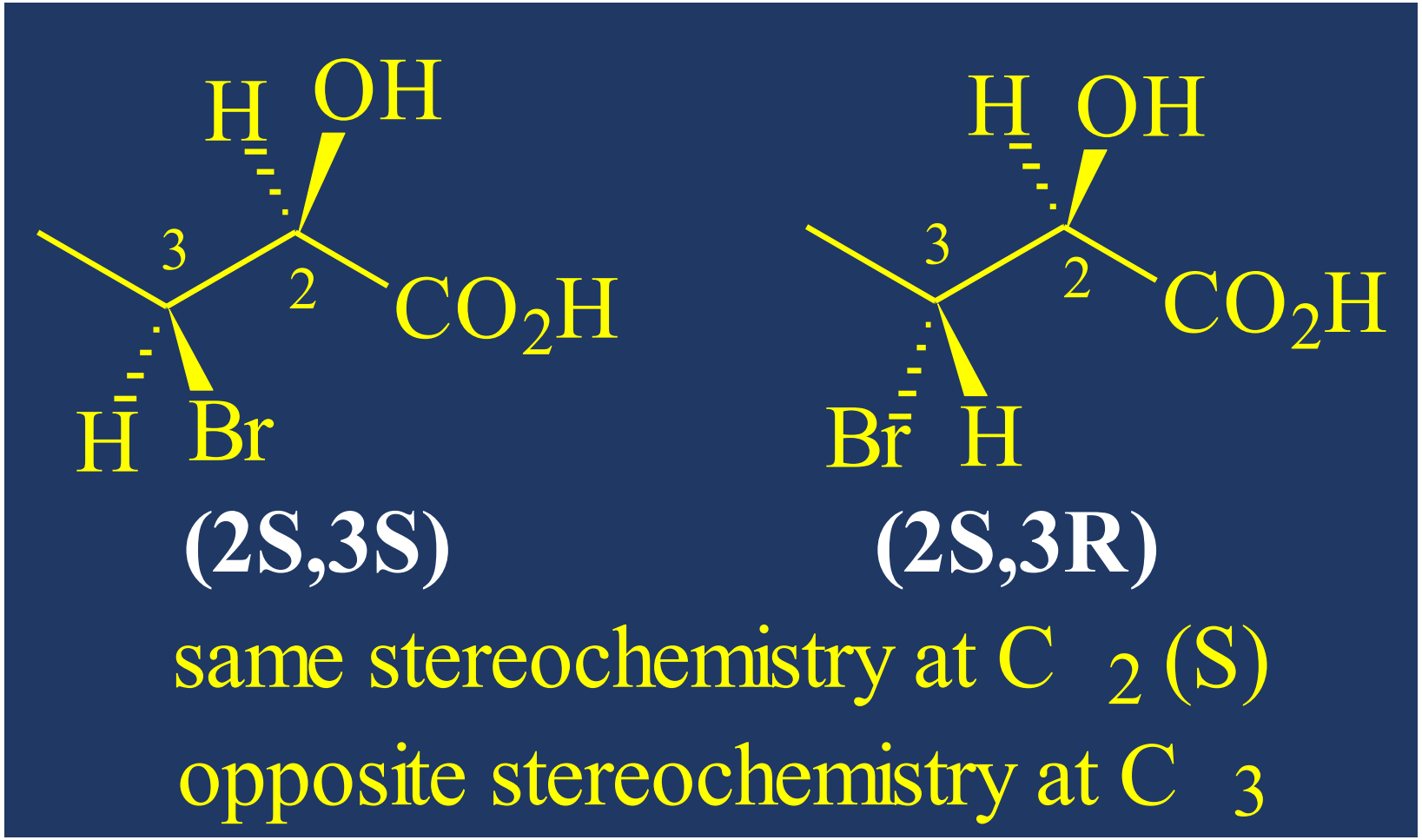
(S)(+) ibuprofen
anti-inflammatory



(R)(-) ibuprofen
80-90% metabolized to (S)(+)

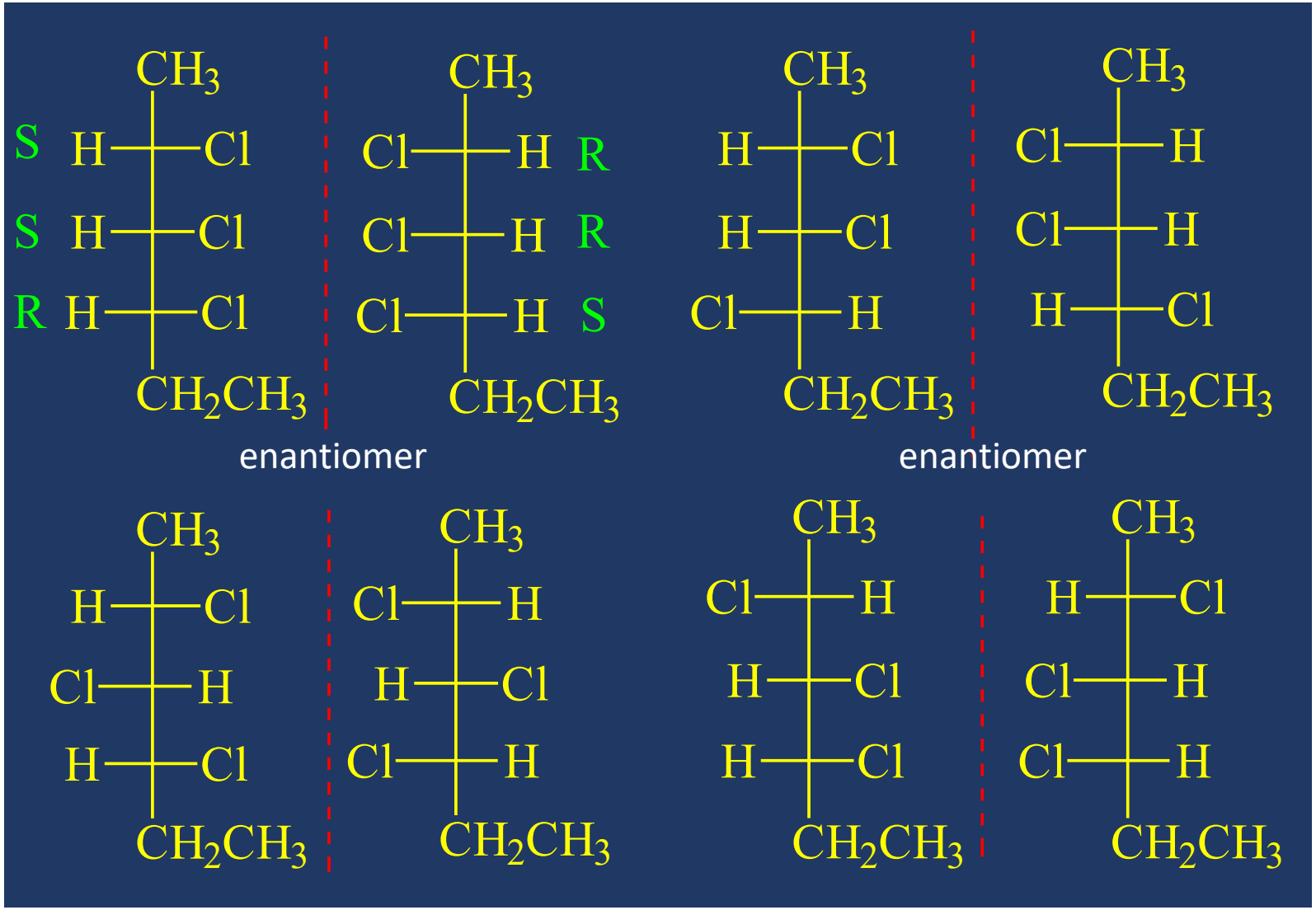
Diastereomers - Not Mirror Images

- This happens when chiral centers are more than two



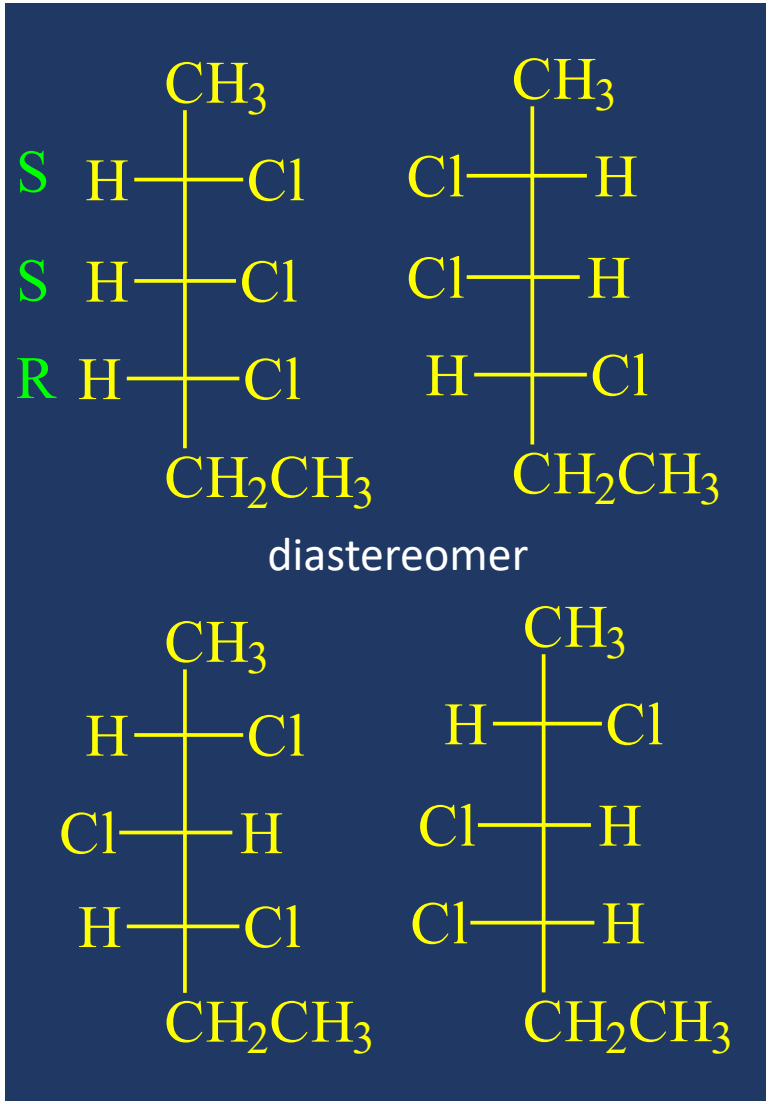
Examples

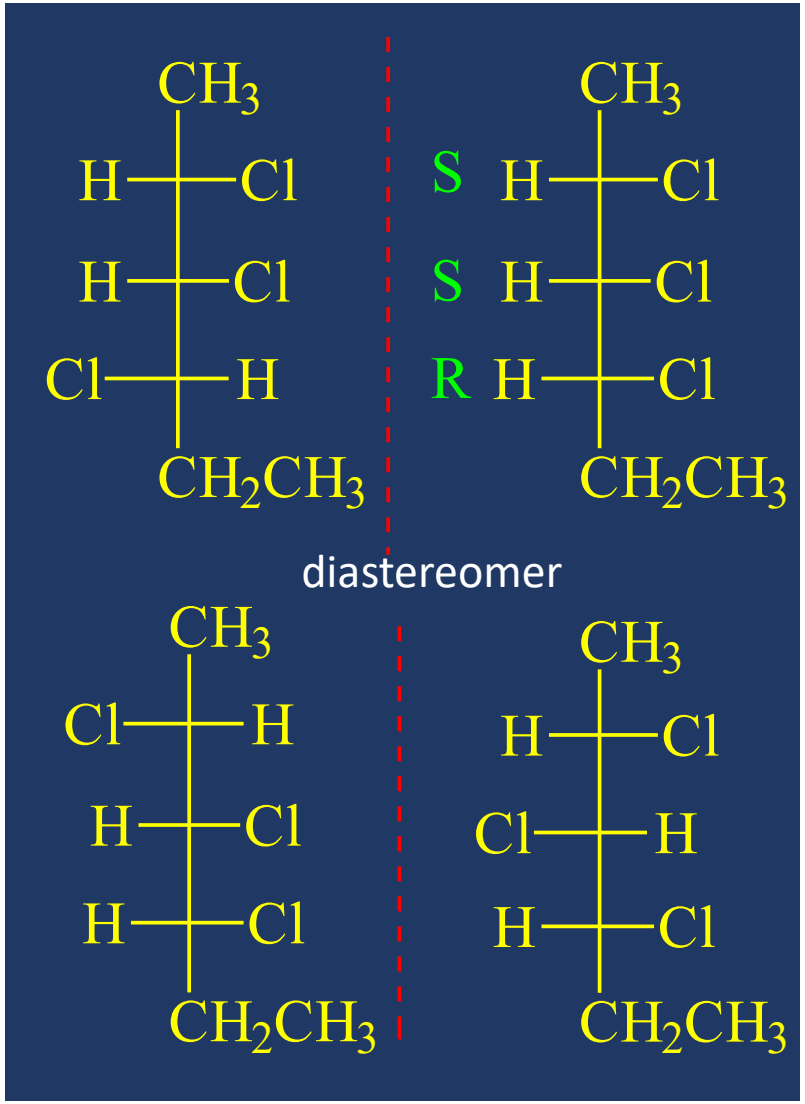
- n = 3; $2^n = 8$



Examples

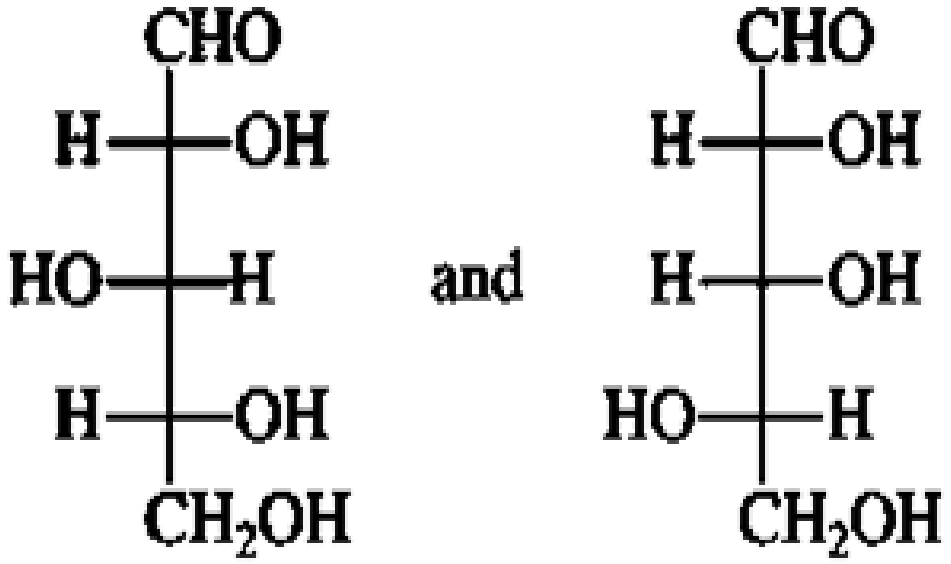
- n = 3; $2^n = 8$

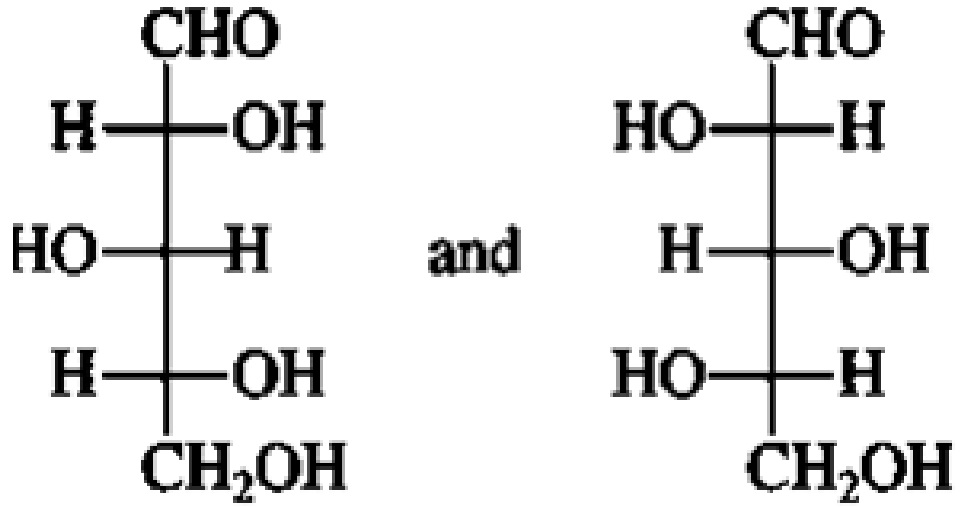




Exercise

Characterize each of the following pairs of structure as enantiomers, diastereomers, or neither enantiomers nor diastereomers.

1. 

2. 

REFERENCES

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